22.6. ACTINIDE CHEMISTRY IN THE ECOSPHERE

22.6.1. Redox properties

Before proceeding to more detailed discussion of the behavior of actinides in the environment, it is useful to review some of their chemical properties. A general discussion of the actinide solution chemistry is given in §14.3; here, the focus is on their behavior in aqueous solutions and primarily in solutions of pH 5–9 which is the pH range of natural waters (e.g. the oceans have pH = 8.2). The actinide ions have an unusually broad range of oxidation states in aqueous solution, from II to VII; The II and VII States are not discussed further as they do not form in ecosystems. Following the normal pattern for polyvalent cations, lower oxidation states are stabilized by more acidic conditions while higher oxidation states are more stable in basic solutions. Of course, this generalization can be negated by other factors, such as complexing, which may cause a reversal of the relative stability of different oxidation states. The greater strength of complexing of An(IV) cations relative to that of An(III) can significantly increase the apparent redox stability of the An(IV) species compared to An(III). The greater tendency to hydrolysis of Pu(IV) causes Pu(III), which is stable in acid solution, to be oxidized to Pu(IV) in neutral media. The disproportionation of Pu(V) is discussed in §14.3.3, where it is pointed out that in the higher pH and very low concentrations of Pu in natural waters, disproportionation is not a factor in the redox behavior of plutonium.

The actinide elements in a particular oxidation state (e.g. Th(IV), U(IV), Pu(IV), Np(IV), Am(IV)) have similar behavior. However, their redox behavior is quite different, as mentioned in Chapter 14. The pH affects this redox behavior significantly as reflected in Figure 22.5 which compares the redox potentials of U, Np and Pu at pH = 0, 8 and 14.

Am (III) is the most stable oxidation state in aqueous solutions while Pu(III) and Np(III) are present under reducing conditions (e.g. anoxic waters). Th(IV) is the common and stable state for that element. U(IV) and Np(IV) do not react with water but are oxidized by O_2 in oxic systems. Pu(IV) is stable at low concentrations in acidic solution, but $Pu(OH)_4$ has a very low solubility product.

NpO_2^+ is stable except at high acidities and high concentrations under which conditions it disproportionates. UO_2^+ and PuO_2^+ increase in stability as the pH is increased. U, Np and Pu form AnO_2^{2+} ions in solution with the stability decreasing in the order U > Pu > Np. UO_2^{2+} is the most stable uranium species in natural waters.

Redox properties are often described by the aid of potential-pH (or Pourbaix) diagrams, Figure 22.6. The shaded area represents typical groundwaters in granitic rock, containing iron minerals; such waters are usually reducing, i.e. have *E*h-values below 0. Natural groundwaters (including oceans, lakes, rivers, etc) fall within the area enclosed by the dashed curve; they can have rather high *E*h values due to atmospheric oxygen, and also be rather alkaline in contact with carbonate rocks. The sloping lines follow the Nernst equation (defined by eqn. (18.6)). We discuss this Figure further in §22.7.1.

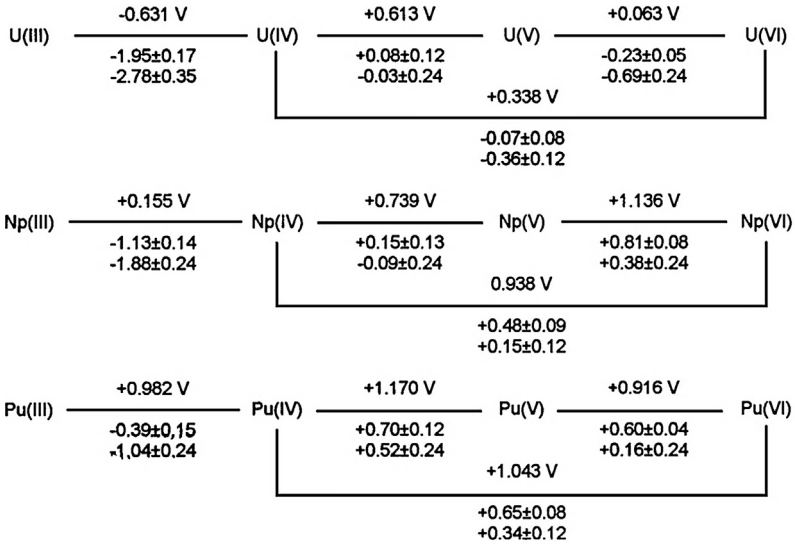


Figure 22.5 Redox potential diagrams of U, Np and Pu; the reduction potentials as listed are for pH values: pH = 0; pH = 8; pH = 14.

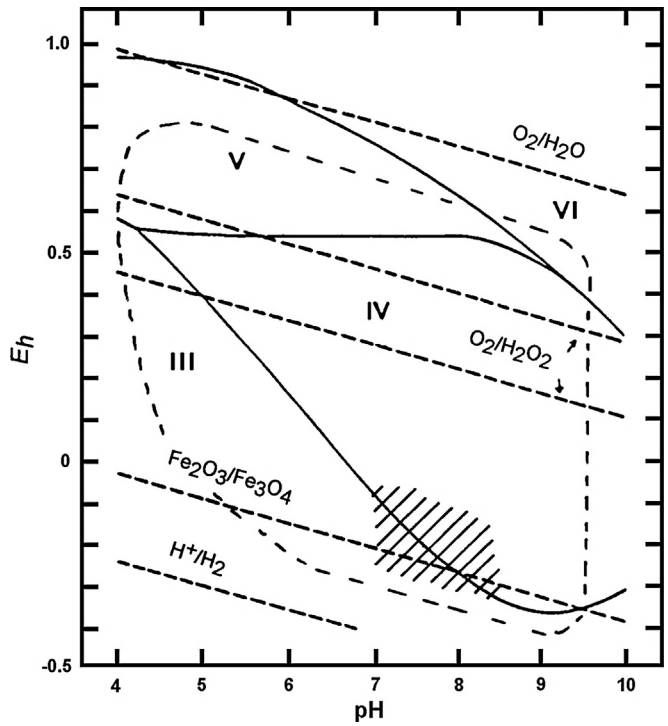


Figure 22.6 Eh-pH (Pourbaix) diagram showing stability areas for Pu(III), Pu(IV), Pu(V) and Pu(VI); at the division line there is an equal concentration of the two oxidation states.

22.6.2. Hydrolysis

Hydrolysis is an important factor in actinide behavior in natural waters as the pH is high enough to result in such reactions as:

$$An^{+n} + mH_2O = An(OH)_m{}^{+(n-m)} + mH^+ \quad (22.1)$$

Since hydrolysis is higly ionic in its nature the main governing factor for the strength of the hydroxide complex is the charge/size ratio. The larger this ratio the stronger the hydrolysis. Therefore, the order of increasing pH for onset of hydrolysis follows the sequence (indeed, in principle all complexation with hard ligands will follow this trend):

$$An^{4+} > AnO_2^{2+} > An^{3+} > AnO_2^{+} \quad (22.2)$$

The variation of the concentration of the free (non- hydrolyzed) cations with pH is shown for the oxidation states of III to VI of Pu in Figure 22.7. These curves are based on estimated values of the hydrolysis constants, but are of sufficient accuracy to indicate the pH values at which hydrolysis becomes significant (e.g. ~6–8 for Pu^{3+}, ≤0 for Pu^{4+}, 9–10 for PuO_2^+ and 4–5 for PuO_2^{2+}.

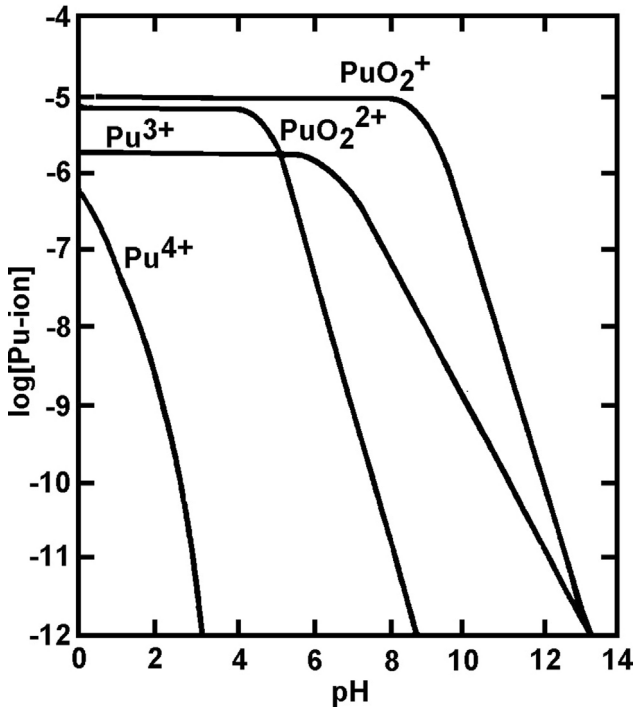


Figure 22.7 Concentration of free plutonium ions at different oxidation states in solutions of different pH, showing the effect of hydrolysis.

The study of plutonium hydrolysis is complicated by the formation of oligomers and polymers once the simple mononuclear hydrolytic species start forming. The relative mono/oligomer concentrations are dependent on the plutonium concentration: e.g. the ratio of Pu present as $(PuO_2)_2(OH)_2^{2+}$ to that as $PuO_2(OH)^+$ is 200 for total $[Pu]_T$ 0.1 M, decreases to 5.6 for 10^{-4} M and is only 0.05 for 10^{-8} M. The hydrolysis of Pu^{4+} can result in the formation of polymers which are very difficult to convert back to simpler species. Generally, such polymerization requires $[Pu]_T > 10^{-6}$ M. However, the irreversibility of polymer formation prevents the destruction of the polymers by dilution of more concentrated hydrolysed solutions to concentrations below 10^{-6} M. Soon after formation, such polymers in solution can be decomposed to simpler species by acidification or by oxidation to Pu(VI). However, as the polymers age, the depolymerization process requires an increasingly vigorous treatment. A reasonable model of the aging involves initial formation of aggregates with hydroxo bridging with conversion over time to structures with oxygen bridging:

The relative percentage of oxygen bridges presumably determines the relative inertness of the polymers. The polymers apparently increase in aggregate size as the pH increases. At pH 4, the polymers are small enough that essentially all of the Pu remains suspended in solution after a week while at pH 5 less than 10% remains in solution and at pH 6, only 0.1% remains.

22.6.3. Solubilities

In marine and natural waters, the limiting solubility is usually associated with either the carbonate or the hydroxide depending on the oxidation state, pH and carbonate concentration. For example for Am^{3+}, the reported value of the solubility product ($\log K_{s0}$, eqn. 18.30) is -26.6 for crystalline $Am(OH)_3(c)$ at very low ionic strength and -22.6 for $Am(OH)(CO_3)(c)$. At pH 6, if $[CO_3^{2-}]$ free $>10^{-12}$ M and at pH 8, if $[CO_3^{2-}]$ free $>10^{-8}$ M , the solubility of Am^{+3} would be expected to be limited by the formation of $Am(OH)(CO_3)$.

Plutonium solubility in marine and natural waters is limited by the formation of $Pu(OH)_4(am)$ (for amorphous) or $PuO_2(c)$ (for crystalline). The K_{s0} of these species is difficult to measure, in part due to the problems of the polymer formation. A measured value for $Pu(OH)_4(am)$ is $\log K_{s0} = -56$. This value puts a limit on the amount of plutonium present, even if Pu(V) or Pu(IV) are the more stable states in the solution phase. Moreover, hydrolyzed Pu(IV) sorbs on colloidal and suspended material, both inorganic and biological.

The strong preference for neptunium to form the NpO_2^+, which has relatively weak complexing and hydrolysis tendency, lead to solubilities as large as 10^{-4} M under many geochemical conditions. However, the reducing environment found at large depths in some granites would make Np(IV) the dominant oxidation state. As with plutonium, the

solubility of neptunium in all oxidation states seems to be limited by the low solubility of $Np(OH)_4$ (am) or NpO_2(c).

In oxic waters, uranium is present as U(VI) and strongly complexes with carbonate; e.g. in sea water, the uranium is present at 10^{-8} M concentration as $UO_2(CO_3)_3^{4-}$. The solubility of uranium in some waters may be limited by the formation of an uranyl silicate species.

22.7. SPECIATION CALCULATIONS

An essential step in the safety analysis of potential waste repositories is the prediction of what chemical species are formed in the actual water. For example, the relatively high solubility of uranium in sea water is due to this strong carbonate complexation which forms $UO_2(CO_3)_3^{4-}$. Figure 22.8 shows the variation of uranyl species in a surface water under normal atmospheric pressure of CO_2 ($p_{CO_2} \sim 3.2 \times 10^{-4}$; $\log[CO_3^{2-}] = 2pH - 18.1 + \log(p_{CO_2})$). These speciation diagrams are calculated from the equilibrium constant for formation of each species plus mass balance equations. In this section we describe the use of equilibrium constants in modeling the speciation in a natural water.

Assume the reactions $M + nX = MX_n$ where X is OH^-, and CO_3^{2-} and $n = 1$ to 3. The equilibrium constants (assuming for simplicity that the chemical activity coefficients all are unity), expressed as β are given by (cf. eqn. (18.31)):

$$\beta_n = [MX_n]/[M]\,[X]^n \tag{22.3}$$

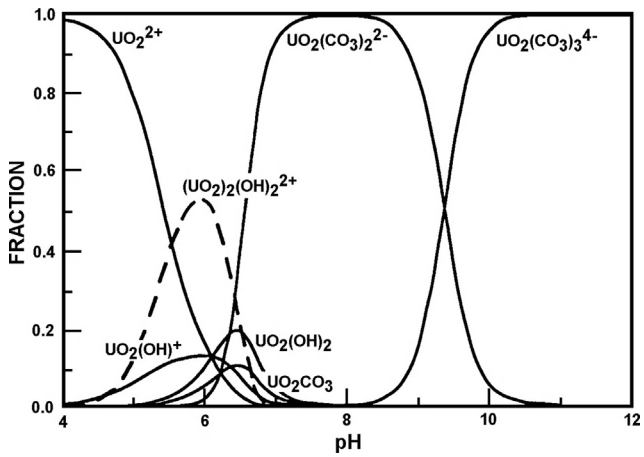


Figure 22.8 Fraction of uranium species in water with natural carbon dioxide content and at different pH, showing hydrolysis and CO_3^{2-} complexation.